GIFTMYBOOK · CHILDREN'S WORKER · TECHNICAL REPORT

Twelve pictures, *one* child.

How we solved character consistency and object consistency across the spreads of an AI-illustrated children's book: the Book Bible, the reference pack, structured verification, and the selection gate.



SAME FACE · SAME OUTFIT · SAME COMPANION · SAME OBJECTS — ON EVERY SPREAD

Scope
Catalog Engine V1.3 · slim illustrator

Engine state
STYLE_VERSION ce-22 · QA_VERSION qa-17

Date
November 2026

01 · SUMMARY

The problem, and the answer in one page

A picture book is twelve illustrations of the same child in the same clothes, with the same friend and the same teddy bear. An image model draws each of those twelve pictures with no memory of the other eleven.

Every spread in our books is a **stateless render**: one independent call to an image model. Left to prose prompts, the model reinvented the child's trousers, shoes and hair, the species of the comfort toy, and the face of the farmer on every page. Each picture looked fine alone; the set did not read as one book.

We solved it by changing *what* the fixed inputs of a render are and *how* a render is accepted:

- **Every identity becomes pixels plus a spec, built once per book.** A character model sheet, an outfit spec read off that sheet, a reference sheet per object and per companion, all collected in a hashed manifest, the **Book Bible**.
- **Every render receives the same reference pack** in a fixed order with fixed labels, and a prompt that states each identity exactly once in structured blocks.
- **Every render is verified against those references**, garment by garment and object by object, by a structured judge with the sheets attached beside the picture, plus deterministic pixel metrics.
- **Candidates are scored and selected**; defects that break the set trigger a bounded repair loop steered only by pinned data.
- **The finished set is checked as a set**: crops of the child, the companion and the objects from all spreads are tiled beside their sheets, and outliers are re-rendered.

0	5	3	3
spread-to-spread chaining. Consistency comes only from fixed inputs.	reference classes in every render's pack: sheet, cover, props, companion, world.	set-level contact gates: child, companion, objects, on top of the world gate.	automatic candidates a spread may buy per run, across every gate (default).

> THE GOVERNING RULE
>
> A stateless render can only be held to fixed inputs, and an invariant that no check enforces is an aspiration. Everything below is that sentence applied to the child, the outfit, the companion and the objects.

02 · DIAGNOSIS

Why twelve stateless renders drift

Before the Book Bible (engine version `ce-9`), the pipeline's doctrine was already right, but three things were wrong with how it was applied to characters and objects:

CAUSE 1

The fixed inputs were words

Clothing rode as one sentence of garment prose; a comfort toy as a quoted noun ("teddy bear"); a companion as "Bea, a farmer". An image model weights images over prose, so every unpictured detail was per-spread freedom.

CAUSE 2

The identity anchor was the wrong kind of image

The approved cover is a scene composition with a title, usually cropping the legs and feet. The garments most prone to drift (hem length, shoes) had no pixels behind them at all.

CAUSE 3

Checks were lenient single-image booleans

QA asked one picture a yes/no question against text ("flag only a clear break"). Nothing compared pixels to pixels. Props and companions had no check at all.

CAUSE 4

Repair was a blind re-roll

One render, a small repair budget, then ship with an advisory. The cache marker replayed the drifted picture forever.

What each trait was held to, before and after

TRAIT	BEFORE	NOW: FIXED INPUT	NOW: CHECK
Face, hair, skin	The cover only; no likeness check	Character model sheet (front, three-quarter, back, head insets) elected once per cover	Identity vs sheet per spread; contact-sheet gate across the set
Clothing	One prose sentence, repeated six times; cropped garments guessed	Per-slot outfit spec derived *from the sheet*, every slot seen, with a hex colour per garment	Slot-by-slot verdict; garment-region colour ΔE vs the spec
Personal object	A quoted noun; nothing pinned, nothing checked	Prop sheet (two angles) plus a structured spec per object	Presence, look, duplication per prop; prop contact gate
Story object	Whatever the beat text implied	Story-object plan: one fixed design, instances, per-spread occurrence and state	Look, required presence, count, `state_match`; critical-family contact check
Companion	A noun on 18% of beats; a new creature or person each spread	Companion sheet plus spec; presence decided by one signal over beat and manuscript	`look_match`, duplication; companion contact gate

WHY NOT CHAIN THE PREVIOUS SPREAD?

Feeding each render the previous spread as a reference looks like the obvious fix. We deleted it: errors compound page by page (photocopy drift), compositions converge, and a single bad page poisons everything after it. Every reference in this design is fixed before the first spread renders and is identical for all twelve.

03 · THE BOOK BIBLE

Identity as pixels plus a spec, built once

Before any spread renders, `buildBookBible` assembles every fixed input for the book. Each component is an **image** the model can copy and a **schema-validated spec** a judge can verify against. Components are elected once (create-if-absent in storage, single winner), so concurrent instances and later retries converge on the same asset.

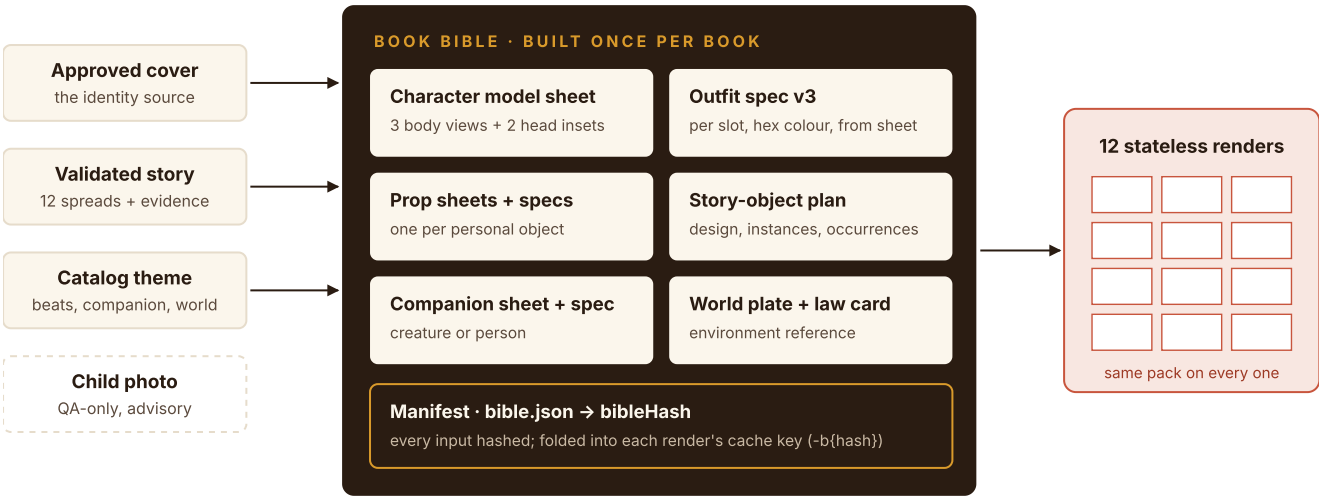


Figure 1. The Book Bible is the only source of identity. Spreads never see each other; they all see the same bible.

The character model sheet

One 16:9 sheet drawn from the approved cover alone: the child full-body front, three-quarter and back in the complete outfit, feet visible, two head insets, a flat grey background, no annotation text. It exists because a cover is a scene, and a scene crops, poses and lights. The sheet shows every garment from every side.

- **Best-of-N election.** Up to three candidates. Each must pass layout and anatomy checks, `cover_identity_matches`, `cover_outfit_matches`, and a cover likeness of at least 0.8. The highest cover likeness wins.
- **The cover is the definition.** The parent approved the cover, so the cover defines face, hair, skin tone, proportions, style and outfit. The raw photo is QA-only: its likeness score is reported and raises a review advisory when low, but it never redesigns the kit.
- **Required by default.** A book that cannot build a sheet fails `identity_kit_failed` instead of silently rendering on the cover alone. A sheet the judge could not verify is never elected, because an elected sheet is pinned for good.
- **Garment lettering is clothing.** A patch, badge or logo on the cover outfit is reproduced exactly and judged under the outfit checks, never as stray text. Otherwise a faithful sheet of an astronaut suit is rejected for its own mission badge.

The outfit spec

A structured spec per slot (top, bottom, footwear, outerwear, accessories), each with colour, cut, the *length words* where drift lives, patterns and graphics, and a `colourHex`. Since `ce-9` it derives from the sheet, where every slot is visible. A slot that would have to be guessed is a derivation failure. The same spec drives the prompt and the check, so two spreads that each pass also match each other.

Prop and companion sheets

One plate per distinct object (two angles, flat background, no text) plus a vision-read structured spec rendered to one inert sentence. The theme companion gets a sheet the same way, cached per theme. Profile strings are data, never instructions: object values are normalised, capped, and quoted.

04 · RENDERING

The reference pack and the structured prompt

Every image call attaches the same pack in a fixed order with fixed labels. The label tells the model what each image is *for*, which matters as much as the image.

#	REFERENCE	ROLE
1	Character sheet	Who the child is and what they wear
2	Approved cover	Identity and style only; never copy its pose or composition
3	Prop sheets	Objects declared on this spread plus carried ones
4	Companion sheet	Only when the companion is on this spread
5	World plate	Environment only, identical on all spreads

The prompt then states each identity **once**, in structured blocks rendered from the bible:

```
CHARACTER    sheet = Reference 1; outfit by slot
PROPS        closed set; each cites its sheet
COMPANION    FIXED LOOK spec; exactly ONE
EMOTION      planned emotion × intensity
ACTION       the catalog beat is the scene
COMPOSITION  shot plan; FACE + BODY lines
WORLD        the theme's law card
```

Design decisions that mattered

- **Say it once.** The old prompt repeated the outfit sentence six times. Repetition did not add weight; it added conflicting phrasings. One block that cites a picture is stronger.
- **Variety is pinned too.** A deterministic shot plan assigns each spread a shot type, staging and placement, seeded by the story fingerprint. Without it, a strong identity reference pulls every spread toward the reference's pose. Consistency of *who* and variety of *how* are pinned by separate mechanisms so they do not fight.
- **The pack survives the safety ladder.** When a prompt is blocked, the sanitized rung strips trigger words from the scene only; pinned blocks stay intact, and the generic-safe rung re-attaches the bible blocks. Before this, a fallback render shipped silently with no outfit and no props. Any render accepted below the first rung now carries an advisory.
- **The cache key is the contract.** The bible hash folds into every render key. Change the cover, a sheet, a spec or an object plan and the affected renders re-key; change nothing and the book replays for free. Locked and lock-less, planned and plan-less renders can never replay each other.

05 · VERIFICATION

Pixels checked against pixels

The per-spread judge (`checkSpreadRenderV2`) receives the render **with the sheets attached beside it** and returns a schema-shaped verdict, not a boolean. It reports identity against the sheet, the outfit garment by garment (`match | mismatch | not_visible`), each object against its sheet (presence, look, duplicated, rendered as text), the companion, the beat's action, the planned emotion, anatomy, and bounding boxes for the child, companion and props.

Each finding maps to a **fixed defect string** classified as blocking or advisory. Only fixed strings and pinned data ever reach a repair prompt; free-form judge prose never does.

CLASS	CHARACTER	OBJECTS AND COMPANION
BLOCKING	Child missing or duplicated; identity, hair or skin differs from the sheet; any visible outfit slot mismatched; extra limbs; body truncated; impossible limb pose; style break	Declared object missing or wrong look; companion wrong look or duplicated; critical story object wrong presence, count or state
ADVISORY	Action or emotion reads differently; hands or face artefacts; face hidden; shot type off	Carried comfort object not visible; undeclared personal object in the scene

Strict fields fail open on a malformed verdict rather than passing silently, and a render cached without a QA marker, or under an older `QA_VERSION`, is re-checked instead of trusted.

Deterministic metrics beside the judge

From the judged bounding boxes, `metrics.js` computes signals that do not depend on a model's opinion: crop and safe-zone rules, and the colour difference (ΔE) between each garment region and the spec's `colourHex`. An optional embedding-based identity score flags set outliers. These shade candidate selection.

06 · SELECTION AND REPAIR

Accepting a render

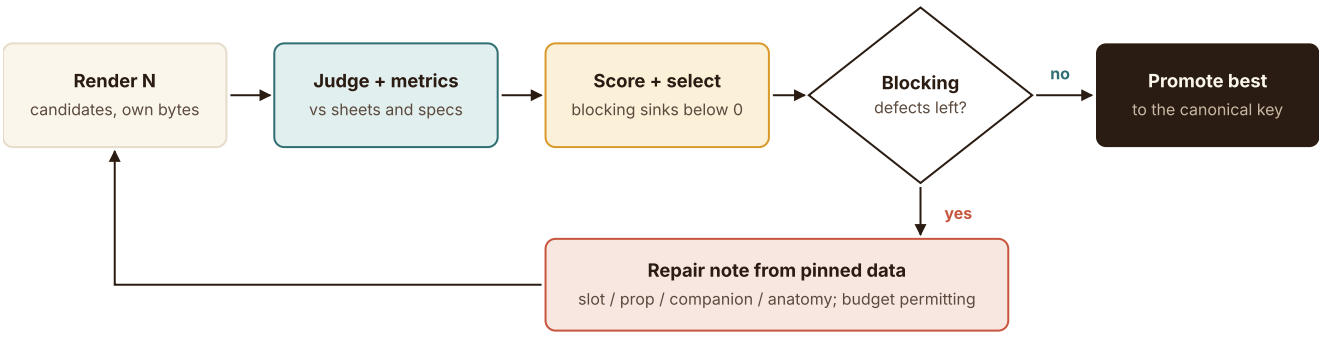


Figure 2. The per-spread acceptance loop. Every automatic candidate, from any gate, draws on one per-spread budget.

- **Scoring.** A blocking defect sinks a candidate below zero. Advisories and metrics shade the rest. An unchecked candidate ranks below any checked one.
- **Every candidate keeps its own bytes.** A rejected repair never overwrites better pixels, and a failure payload lists exactly what was scored.

- **A repair is adopted only when it scores higher.** An unchecked repair (checker outage mid-loop) never replaces a render whose defects are known. A set-gate re-render with *more* blocking defects than the flagged render is discarded and the original bytes restored.
- **One budget.** Base candidates, repairs, set-gate re-renders and lettering recovery all spend from `CATALOG_RENDER_BUDGET_PER_SPREAD` (default 3). Cost is bounded per spread, whatever combination of gates fires.

> SHIP POLICY
>
> The book finishes. When the budget is spent, a spread ships its best candidate with every finding on record in the callback, and the page is named for proofing. The one thing no policy can carry is a missing image, which gets bounded fresh render rounds. A strict mode (`CATALOG_SHIP_ON_EXHAUSTION=0`) turns every blocking residual back into a hard stop with scored candidates attached, and an admin can promote a candidate by hand.

07 · SET GATES

Judging the book as a book

Twelve spreads that each pass against the sheet can still disagree with each other inside the tolerance. So after the per-spread loop, the run is judged as a set.

WORLD GATE

One multi-image check across the run

A closed vocabulary of set breaks: palette, era, materials, magic behaviour, plus `character_rendering` (the child reads as a different age, proportions, outfit or hair) and `composition_duplicate`.

CONTACT-SHEET GATE · CHILD

Crops tiled beside the model sheet

At thumbnail scale, garments are invisible. The gate crops the child from every spread using the judged bounding box and tiles the crops next to the sheet in one image, so hem lengths and shoes are legible.

CONTACT-SHEET GATE · OBJECTS

Per-prop crops beside each prop sheet

Flags `prop_rendering`. A repair cites the prop sheet's index in the re-render's own pack. Critical story-object families get a final contact check on the selected artwork.

CONTACT-SHEET GATE · COMPANION

Every spread that expected the companion

Flags `companion_rendering`. This is the set-level view that two individually passing spreads never get.

Flagged **fresh** spreads get one corrective re-render through the full per-spread path, under a per-gate cap. Replayed cached renders are comparison references only and are never re-rendered by a gate, because their bytes may already be approved elsewhere. Gate findings ship as advisories, and each gate's verdict rides the completion callback.

08 · OBJECTS

Three kinds of object, three policies

"Object consistency" turned out to be three different problems. Treating them as one caused both drift and over-correction.

KIND	EXAMPLE	POLICY
Personal object	The child's comfort toy, from the profile	A prop sheet and spec. Introduced on its declared evidence spread (missing there is blocking), then **carried through** every later spread as a small, visually subdued continuity prop (not visible later is advisory). Food, place and interest details stay pinned to their declared spreads only.
Story object	A route marker, a key, a map, a vehicle	A **story-object plan**: one fixed design per family, explicit instances, and a per-spread occurrence schedule with physical state. Plot objects do not inherit the "small and decorative" rule.
Everything else	Invented toys, gadgets, trinkets	The personal-object set is **closed**. The prompt forbids objects beyond the declared set and the beat's own needs; the judge reports `undeclared_object`.

The story-object plan

`storyObjects.js` reads the final manuscript, the catalog beats, the theme and any authored definitions together, and resolves object families, aliases, instances, per-spread state and story-criticality.

- **Grounded evidence.** The planner selects evidence by numbered source passage; the worker resolves those ids to exact substrings of the manuscript or beat. This removes model transcription errors, and an explicit mention cannot silently lose its occurrence.
- **A fixed design.** Shape, material, colours, scale, features. Authored catalog definitions cannot be omitted or rewritten. The first one: a knee-high wooden post with one orange stripe, shared by every route marker in a safari book. The displaced third marker keeps its instance id.
- **Identity is separate from visibility.** Groups are intentional. A moved, damaged or repaired instance keeps its identity; only its state changes. The judge verifies `state_match` including stated clues and spatial relationships.
- **Field-scoped completion.** Once identities validate, the plan is frozen and only defective fields are completed, under a bounded budget. A partial plan is never elected.
- **Elected and hashed.** The plan is elected create-if-absent by input hash, and its hash enters the bible and every spread's QA record. A changed plan cannot reuse older approval.

References that heal themselves

A reference sheet is itself an AI image and can fail its own check. Because every root on that path is content-fingerprinted, a regeneration used to replay the saved rejection forever. The path now heals in a fixed order:

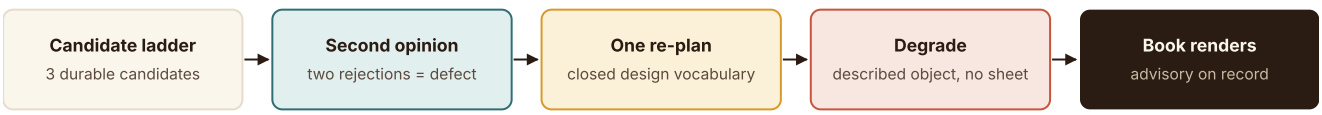


Figure 3. Reference auto-heal. A degraded critical object keeps presence, count and state blocking; only its look becomes advisory.

An explicit regeneration opens fresh candidate slots under a retry namespace, while a plain resume replays saved work. The same rule applies to the character sheet: if the image provider refuses to draw it on every rung of the prompt ladder, the book renders on the approved cover alone with a loud advisory rather than pausing on a review that could only replay the same refusal.

09 · SECONDARY CHARACTERS

The companion is a Bible identity

A farm book shipped with Farmer Bea as a different woman on two spreads. The investigation found two causes, both general, and the fix (`ce-19`) treats every named recurring companion exactly like the child.

CAUSE

Person companions had no pixels

Human companions were excluded from the companion sheet because the renderer forbade inventing adult faces. Two themes (38 books) rendered their guide as a bare noun: no sheet, no spec, no look check, no set gate.

CAUSE

The presence signal never fired

World-name masks were raw substring splits. The farm theme's display name "Farm" is a substring of "Farmer", so "Farmer Bea" became "er Bea" before the name was looked for. No companion line, no block, no check, on any spread.

What changed

- **One presence signal.** `companionOnSpread` reads the beat *and* the manuscript text: the companion's name as a case-sensitive whole word ("Patch" the parrot never fires on "a patch of mud"), or the full type phrase. That one signal gates the scene line, the sheet in the pack, the COMPANION block and the QA check. Masks are whole-word.
- **One kind decision.** A shared module decides person versus creature for the sheet builder, the renderer, the judge and the contact gate. Previously two regexes disagreed.
- **A secondary character sheet.** A person companion gets a model sheet (one fictional person, full-body, two views) and a character spec: apparent age and build from a closed vocabulary, skin tone, hair, face notes, outfit by garment, dominant hex colours.
- **The spec rides the prompt.** The companion's spec sentence is stated as FIXED LOOK with an "exactly ONE" rule. Before, it was derived, hashed and never used.
- **Checked like the child.** `look_match` by face, age, hair, skin, build and outfit; a duplication check; and the companion contact-sheet gate across the set.

THE GENERAL LESSON

A blanket safety rule ("no invented human faces in the background") had silently contradicted a product requirement on 38 books. The fix was to make the exception explicit and bounded: one named fictional person, pinned by a sheet, allowed by the COMPANION block.

10 · EVOLUTION

How we got here

The solution was reached in steps. The early steps each fixed a symptom with more prompt text; the turning point was `ce-9`, which replaced words with pixels and booleans with structured verdicts.

- **pre** Previous-spread chaining deleted as the source of photocopy drift. Consistency must come from fixed inputs.
- **ce-5** World-law card on every prompt, a fixed per-theme world plate, and the first set-level gate.
- **ce-6** Carry-through props: the comfort object persists after its evidence spread.
- **ce-7** Outfit lock: a per-anchor clothing spec read from the cover, pinned on every render.
- **ce-8** Structured per-slot outfit spec, outfit QA against it, and the deterministic shot plan for variety.
- **ce-9** **The Book Bible and the selection gate.** Character sheet, sheet-derived outfit spec, prop and companion sheets, reference pack, structured judge with sheets attached, candidate scoring, contact-sheet gate.
- **ce-10** Closed personal-object set, subdued carried props, garment patterns in the outfit check, face visibility.
- **ce-11** Companion presence pinned by beat and manuscript, so mid-book appearances get the sheet and the check.
- **ce-19** Secondary characters as Bible identities: person companion sheets, specs and the companion contact gate.
- **ce-20** The whole body: truncated bodies and impossible limb poses become blocking, beyond limb counting.
- **so-1** Story-object plan: fixed designs, instances, per-spread occurrence and state for plot objects.
- **heal** Self-healing references; garment lettering judged as clothing; refusal fallback for the sheet.
- **ce-22** The approved cover defines the identity; the photo becomes an advisory signal only.

11 · LESSONS

What generalises

1. Pin pixels, not adjectives

If an identity matters, give the model a picture of it and a judge a spec of it. An unspecified garment or an unpictured toy is per-spread freedom.

2. One spec for prompt and check

When the same pinned spec drives the render and the gate, they cannot disagree, and spreads that each pass also match each other.

3. A gate can only reject

Rejection without a better input just burns budget. Every gate here is paired with a fixed input that makes the right result likely on the first render.

4. Compare at the scale of the defect

A shoe is invisible in a twelve-image thumbnail grid. Crop the subject and tile the crops beside the reference.

5. Repairs speak a closed vocabulary

Only fixed defect strings and pinned data reach a repair prompt. Judge prose in a prompt is an injection path and a drift source.

6. Never adopt worse, never trust unchecked

Candidates keep their own bytes, repairs must score higher, and a render without a current QA marker is re-checked.

7. Fixed references need their own recovery

A pinned asset that fails its check will replay that failure forever unless there is an ordered path: second opinion, re-plan, degrade, advise.

8. Version what changes pixels

Every input is hashed into the cache key, so approved art replays byte-for-byte and only what a change actually touches re-renders.

Operational reference

MODULE	RESPONSIBILITY
`illustrator/bible/index.js`	Builds the Book Bible and its manifest
`bible/characterSheet.js`	Character model sheet: candidates, judge, election
`outfitLock.js`	Per-slot outfit spec derived from the sheet
`bible/propSheet.js`	Prop and companion sheets and specs; auto-heal
`storyObjects.js`, `storyObjectRepair.js`	Story-object plan, grounding, field-scoped completion
`scenes.js`	Scene prompt, `companionOnSpread`, continuity props
`spreadQa.js`, `metrics.js`, `select.js`	Structured verdict, deterministic signals, scoring
`contactSheet.js`	Child, prop and companion contact-sheet gates

SWITCH (DEFAULT)	EFFECT WHEN CHANGED
`CATALOG_CHARACTER_SHEET` **(on)**	`0`: renders anchor on the cover alone
`CATALOG_SHEET_REQUIRED` **(on)**	`0`: a failed sheet degrades to an advisory
`CATALOG_PROP_SHEETS` **(on)**	`0`: objects ride as described objects
`CATALOG_HUMAN_COMPANION_SHEET` **(on)**	`0`: person companions ride as nouns
`CATALOG_PROP_CONTINUITY` **(on)**	`0`: no carried comfort object
`CATALOG_CONTACT_QA` **(on)**	`0`: skip the contact-sheet gates
`CATALOG_REFERENCE_AUTOHEAL` **(on)**	`0`: reference failures pause the book
`CATALOG_SHIP_ON_EXHAUSTION` **(on)**	`0`: blocking residuals stop the run
`CATALOG_RENDER_BUDGET_PER_SPREAD` **(3)**	Automatic candidates per spread across all gates